

Assessing racial, ethnic, and nativity disparities in US cancer mortality using a new integrated platform

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Abstract

Background: Foreign-born populations in the United States have markedly increased, yet cancer trends remain unexplored. Survey-based Population-Adjusted Rate Calculator (SPARC) is a new tool for evaluating nativity differences in cancer mortality.

Methods: Using SPARC, we calculated 3-year (2016–2018) age-adjusted mortality rates and rate ratios for common cancers by sex, age group, race and ethnicity, and nativity. Trends by nativity were examined for the first time for 2006–2018. Traditional cancer statistics draw populations from decennial censuses. However, nativity-stratified populations are from the American Community Surveys, thus involve sampling errors. To rectify this, SPARC employed bias-corrected estimators. Death counts came from the National Vital Statistics System.

Results: Age-adjusted mortality rates were higher among US-born populations across nearly all cancer types, with the largest US-born, foreign-born difference observed in lung cancer among Black women (rate ratio = 3.67, 95% confidence interval [CI] = 3.37 to 4.00). The well-documented White–Black differences in breast cancer mortality existed mainly among US-born women. For all cancers combined, descending trends were more accelerated for US-born compared with foreign-born individuals in all race and ethnicity groups with changes ranging from –2.6% per year in US-born Black men to stable (statistically nonsignificant) among foreign-born Black women. Pancreas and liver cancers were exceptions with increasing, stable, or decreasing trends depending on nativity and race and ethnicity. Notably, foreign-born Black men and foreign-born Hispanic men did not show a favorable decline in colorectal cancer mortality.

Conclusions: Although all groups show beneficial cancer mortality trends, those with higher rates in 2006 have experienced sharper declines. Persistent disparities between US-born and foreign-born individuals, especially among Black people, necessitate further investigation.

As of 2023, the United States has more than 334 million people, and 15% of them are foreign-born (1). Their impact in the US melting pot is exceptional because of their social and intellectual contributions, as well as their diverse lifestyles, rich cultures, different diets, and distinct patterns of familial cohesion and support (2). Naturally, such diversity could lead to different cancer mortality profiles, which are important to characterize on 2 levels: first, to understand these populations' disease burden and meet their health needs and, second, to foster a full understanding of the etiology of cancer. On a population basis, birthplace may serve as a proxy for many characteristics tied to access to health care, environmental influences, and lifestyle factors, particularly during the formative years (3). Moreover, birthplace is an objective variable, more so than race and/or ethnicity, and is readily available in mortality data and useful for inclusive population-based analyses.

Past studies have shown striking differences in cancer mortality between US-born and foreign-born populations. These

differences are partially driven by differential racial and ethnic composition in that US-born individuals are predominantly White followed by Black, whereas foreign-born individuals are mainly Hispanic and Latinx followed by Asian (1). Among Black and Hispanic individuals, the US-born have been shown to have higher mortality rates than their foreign-born counterparts (4–6) for most cancer sites, which are largely attributed to high prevalence of crucial risk factors among the US-born populations including smoking, obesity, alcohol, and drug use, and specific reproductive patterns (7). Notable examples are lung cancer mortality rates being 3–4 times higher in US-born Black populations than in foreign-born Black populations (5,8,9), and liver cancer mortality rates in men being twice as high in US-born Hispanic populations compared with foreign-born Hispanic populations (10–12). For Asian and Pacific Islander populations, nativity-specific mortality studies are scarce, and results are mixed. One study shows higher rates among the US-born Asian and Pacific Islander individuals (6), whereas another suggests lower rates in

cancers that are infection-, diet-, and tobacco-related or screen detected in this group (13).

Therefore, aggregating the US-born and the foreign-born populations into one group limits our ability to identify subpopulations for achieving health equity. All racial and ethnic subpopulations, especially Black, Hispanic, and Asian and Pacific Islander, should be studied by distinct birthplace groups whenever possible. Over time, cancer patterns by nativity are dynamic with changing demographic weights of racial and ethnic groups, and trends by cancer site are distinct because of shifting incidence, improved detection, and cancer treatment. The study of nativity-specific trends fills an important gap in descriptive epidemiology, which could unveil key insights into the causes and control of cancer.

To facilitate characterizing these epidemiologic disparities, the National Cancer Institute (NCI) developed the Survey-Based Population-Adjusted Rate Calculator (SPARC) (14), a web-based tool for calculating time series of nativity-specific mortality rates at the national and state levels. This tool is equipped with authoritative data about mortality and population as well as novel statistical methods specifically addressing arising challenges because of sampling errors in nativity-specific populations. The lack of census-based populations has led to the use of sample-based estimates as denominators in the calculation of cancer rates. However, sampling errors bias the estimation of rates and comparisons of rates for disparity evaluation (15,16). Using aggregated data to mitigate the impact is a common approach (17), but it comes with the cost of losing temporal-spatial granularities crucial for understanding the nuances of trends and patterns. This new tool enables easy analyses of rigorous trend statistics for the first time. The democratization of complex rate analysis has an important potential to contribute to widespread and effective analysis of cancer data and to facilitate evidence-based policy making for effective resource allocation.

In this article, we illustrate SPARC by examining recent 3-year US mortality rates for the most common causes of cancer death, segmented by age, sex, race and ethnicity, and nativity from 2016 to 2018. Furthermore, we present an inaugural analysis of cancer-specific mortality trends by nativity over a 12-year timeline from 2006 to 2018.

Methods

Mortality data

Annual cancer-specific mortality counts from 2006 to 2018 for all US states and District of Columbia were retrieved from the National Vital Statistics Systems (18). These counts are stratified by state, 5-year age grouping (0-4, 4-9, 10-14, ..., 80-84 years, and 85 years and older), sex (female or male), bridged single race and ethnicity (non-Hispanic White, non-Hispanic Black, non-Hispanic Asian and Pacific Islander, non-Hispanic American Indian and Alaska Native, and Hispanic), causes of death recode (19), and nativity (US-born vs foreign-born), where the foreign-born population is defined as anyone who is not a US citizen at birth (20) and the US-born is everyone else, including those who are born in US territories.

Deaths with missing age (<0.01%) or race and ethnicity (0.27%) were excluded. Nativity is highly complete with missing data no greater than 2% for most states and death years, except for Georgia, where more than 80% is missing in 2008-2009 during Georgia's transition to electronic capture of death certificate reporting. For the trend analysis that spans in years from 2006 to

2018, deaths from Georgia were excluded to ensure coverage comparability over time.

Population denominators

Annual populations and standard errors by the same variables as defined in the mortality data were estimated from 2006 to 2018 1-year American Community Survey (ACS) samples extracted from the University of Minnesota's Integrated Public Use Microdata Series (1,21). Small proportions of missing data in race and ethnicity (<1%) or nativity data (<1%) in ACS samples were multiply imputed (22) to overcome underestimation biases because of excluding missing data. Sampling weights and stratification were incorporated in the estimation of populations and standard errors (23).

Inference methods of age-adjusted mortality rates and rate ratios

The age-adjusted mortality rate is a weighted sum of ratios of deaths and populations in different age groups. Unlike the usual census-based population denominators, which are typically assumed to be free of error when used to estimate age-adjusted mortality rates, ACS-based populations are subject to sampling errors. Sampling errors cause biases in estimating age-adjusted mortality rates (upward bias), standard errors of age-adjusted mortality rates (downward bias), and ratios of age-adjusted mortality rates in comparing two subpopulations. An estimate is unbiased if a positive deviation from the true parameter (because of random errors) is always matched by a negative deviation over a large number of random samples; thus the estimate equals the parameter on average. However, in the case of estimating age-adjusted mortality rates, sampling errors are inversely related to age-adjusted mortality rates. Even though sampling errors can cancel each other, the resulting random deviations in age-adjusted mortality rates tend to be positive on average, thus leading to an upward bias. The presence of sampling errors also increases statistical variability. These impacts are greater with larger sampling errors, which are likely to occur in small demographic and geographic groups. This differential impact across subpopulations of varying sizes further causes bias in estimating rate ratios, an important measure for assessing disparities. Newly developed bias-corrected methods (15,16), as incorporated in SPARC, were used to account for these biases.

Data analysis

Bias-corrected 3-year (2016-2018) US mortality age-adjusted mortality rates (per 100 000 people) and standard errors were computed using SPARC for all cancers combined and the 5 most common causes of cancer deaths (ie, female breast, prostate, lung [including bronchus], colorectum, and pancreas, as well as liver and intrahepatic bile duct); by sex; race and ethnicity (Hispanic, non-Hispanic White, non-Hispanic Black, and non-Hispanic Asian and Pacific Islander); nativity; and age group (younger than 50 years vs 50 years and older). For simplicity, we refer to non-Hispanic White, non-Hispanic Black, and non-Hispanic Asian and Pacific Islander groups without the *non-Hispanic* prefix with the understanding that these are mutually exclusive from the Hispanic group, as they can be of any race. Rate ratios were used for comparisons of the US-born with the foreign-born within each race and ethnicity. To examine the trends, we exported a comma-separated values file of annual age-adjusted mortality rate for the period of 2006-2018 from SPARC and imported them to NCI's Joinpoint software to fit Joinpoint linear regression models (24) by sex, race and ethnicity,

Table 1. Age-adjusted mortality rates per 100 000 persons and rate ratios of comparing US-born with foreign-born individuals by sex, cancer site, race and ethnicity, and age group, 2016–2018, United States^a

Sex	Cancer site	Race and ethnicity	All ages			Aged younger than 50 years			Aged 50 years and older		
			Age-adjusted mortality rate		Rate ratio (reference: foreign-born)	Age-adjusted mortality rates		Rate ratio (reference: foreign-born)	Age-adjusted mortality rate		Rate ratio (reference: foreign-born)
			US-born	Foreign-born	Rate ratio (95% CI) ^a	US-born	Foreign-born	Rate ratio (95% CI)	US-born	Foreign-born	Rate ratio (95% CI)
Male	All cancers	Non-Hispanic White	187.10	158.58	1.18 (1.16 to 1.20)	14.60	11.42	1.28 (1.20 to 1.37)	638.83	543.94	1.17 (1.16 to 1.19)
		Non-Hispanic Black	233.03	129.90	1.79 (1.72 to 1.87)	17.36	12.83	1.35 (1.24 to 1.48)	797.79	436.44	1.83 (1.75 to 1.91)
		Non-Hispanic Asian and Pacific Islander	114.15	115.50	0.99 (0.94 to 1.04)	9.93	10.78	0.92 (0.82 to 1.03)	387.05	389.72	0.99 (0.94 to 1.04)
	Prostate	Hispanic	143.68	125.16	1.15 (1.12 to 1.18)	11.73	10.60	1.11 (1.04 to 1.17)	489.21	425.16	1.15 (1.12 to 1.18)
		Non-Hispanic White	18.09	16.50	1.10 (1.06 to 1.14)	0.08	0.07	1.18 (0.58 to 2.42)	65.26	59.53	1.10 (1.06 to 1.14)
		Non-Hispanic Black	39.54	30.32	1.30 (1.21 to 1.40)	0.25	0.13	1.99 (0.92 to 4.31)	142.42	109.37	1.30 (1.21 to 1.40)
	Lung and bronchus	Non-Hispanic Asian and Pacific Islander	10.78	8.96	1.20 (1.06 to 1.37)	0.03	0.05	0.72 (0.09 to 5.79)	38.90	32.30	1.21 (1.06 to 1.37)
		Hispanic	15.33	16.32	0.94 (0.88 to 1.00)	0.09	0.09	1.02 (0.57 to 1.82)	55.23	58.84	0.94 (0.88 to 1.00)
		Non-Hispanic White	47.37	33.86	1.40 (1.36 to 1.44)	1.73	0.87	2.00 (1.64 to 2.44)	166.87	120.25	1.39 (1.35 to 1.42)
		Non-Hispanic Black	58.01	16.83	3.44 (3.18 to 3.72)	2.15	0.83	2.59 (1.92 to 3.49)	204.27	58.74	3.47 (3.21 to 3.76)
Female	Lung and bronchus	Non-Hispanic Asian and Pacific Islander	26.59	27.66	0.96 (0.89 to 1.04)	1.07	1.03	1.03 (0.72 to 1.48)	93.41	97.37	0.96 (0.88 to 1.04)
		Hispanic	22.53	21.32	1.06 (1.01 to 1.11)	0.66	0.53	1.24 (0.95 to 1.60)	79.81	75.76	1.05 (1.00 to 1.11)
		Non-Hispanic White	15.73	15.52	1.01 (0.98 to 1.05)	2.18	1.49	1.46 (1.25 to 1.70)	51.22	52.23	0.98 (0.95 to 1.02)
		Non-Hispanic Black	24.01	13.67	1.75 (1.61 to 1.91)	3.10	1.44	2.15 (1.71 to 2.71)	78.78	45.70	1.72 (1.57 to 1.89)
	Colon and rectum	Non-Hispanic Asian and Pacific Islander	11.36	11.15	1.02 (0.91 to 1.14)	1.54	1.41	1.09 (0.82 to 1.47)	37.09	36.67	1.01 (0.90 to 1.14)
		Hispanic	16.47	11.95	1.38 (1.31 to 1.45)	1.74	1.21	1.44 (1.25 to 1.65)	55.05	40.09	1.37 (1.30 to 1.45)
		Non-Hispanic White	13.06	12.76	1.02 (0.98 to 1.06)	0.84	0.75	1.12 (0.90 to 1.40)	45.06	44.21	1.02 (0.98 to 1.06)
		Non-Hispanic Black	16.05	10.42	1.54 (1.40 to 1.70)	1.17	0.62	1.88 (1.32 to 2.66)	55.04	36.08	1.52 (1.38 to 1.69)
	Pancreas	Non-Hispanic Asian and Pacific Islander	10.38	7.88	1.32 (1.17 to 1.49)	0.48	0.52	0.91 (0.55 to 1.52)	36.30	27.15	1.34 (1.18 to 1.52)
		Hispanic	9.76	9.46	1.03 (0.97 to 1.10)	0.70	0.52	1.35 (1.07 to 1.69)	33.49	32.85	1.02 (0.95 to 1.09)
Liver and intrahepatic bile duct	Non-Hispanic White	8.43	7.55	1.12 (1.06 to 1.18)	0.49	0.44	1.12 (0.77 to 1.62)	29.22	26.16	1.12 (1.06 to 1.18)	
	Non-Hispanic Black	13.60	8.34	1.63 (1.47 to 1.80)	0.61	2.31	0.27 (0.21 to 0.33)	47.60	24.13	1.97 (1.76 to 2.21)	
	Non-Hispanic Asian and Pacific Islander	8.97	13.59	0.66 (0.59 to 0.74)	0.73	1.58	0.47 (0.31 to 0.69)	30.53	45.04	0.68 (0.60 to 0.77)	
	Hispanic	18.41	9.77	1.88 (1.79 to 1.99)	0.84	0.53	1.60 (1.29 to 1.97)	64.40	33.98	1.90 (1.79 to 2.00)	
(continued)											

(continued)

Table 1. (continued)

Sex	Cancer site	Race and ethnicity	All ages			Aged younger than 50 years			Aged 50 years and older		
			Age-adjusted mortality rate		Rate ratio (reference: foreign-born)	Age-adjusted mortality rates		Rate ratio (reference: foreign-born)	Age-adjusted mortality rate		Rate ratio (reference: foreign-born)
			US-born	Foreign-born		US-born	Foreign-born		US-born	Foreign-born	
Female	All cancers	Non-Hispanic White	135.95	117.98	1.15 (1.14 to 1.17)	16.48	14.72	1.12 (1.06 to 1.19)	448.78	388.38	1.16 (1.14 to 1.17)
		Non-Hispanic Black	157.67	102.70	1.53 (1.49 to 1.58)	22.13	16.88	1.31 (1.22 to 1.41)	512.57	327.44	1.56 (1.51 to 1.62)
		Non-Hispanic Asian and Pacific Islander	84.84	82.19	1.03 (0.99 to 1.08)	11.93	11.79	1.01 (0.91 to 1.12)	275.76	266.54	1.04 (0.99 to 1.08)
		Hispanic	96.98	90.54	1.07 (1.05 to 1.10)	14.26	13.19	1.08 (1.03 to 1.14)	313.60	293.11	1.07 (1.05 to 1.10)
	Breast	Non-Hispanic White	19.84	19.77	1.00 (0.97 to 1.03)	3.92	4.28	0.91 (0.83 to 1.00)	61.52	60.34	1.02 (0.99 to 1.05)
		Non-Hispanic Black	28.75	20.33	1.41 (1.34 to 1.49)	7.68	5.78	1.33 (1.19 to 1.49)	83.94	58.43	1.44 (1.35 to 1.53)
		Non-Hispanic Asian and Pacific Islander	13.48	11.21	1.20 (1.10 to 1.32)	3.26	2.84	1.15 (0.94 to 1.40)	40.24	33.12	1.22 (1.10 to 1.34)
		Hispanic	14.82	12.55	1.18 (1.13 to 1.23)	3.31	3.05	1.09 (0.99 to 1.20)	44.96	37.45	1.20 (1.14 to 1.26)
	Lung and bronchus	Non-Hispanic White	34.73	21.09	1.65 (1.60 to 1.69)	1.65	0.78	2.11 (1.72 to 2.60)	121.36	74.27	1.63 (1.59 to 1.68)
		Non-Hispanic Black	31.61	8.60	3.67 (3.37 to 4.00)	1.21	0.80	1.52 (1.10 to 2.09)	111.23	29.04	3.83 (3.50 to 4.18)
		Non-Hispanic Asian and Pacific Islander	17.38	15.28	1.14 (1.05 to 1.24)	0.46	1.17	0.39 (0.24 to 0.64)	61.70	52.22	1.18 (1.08 to 1.29)
		Hispanic	13.17	10.48	1.26 (1.20 to 1.32)	0.59	0.53	1.13 (0.90 to 1.41)	46.11	36.53	1.26 (1.20 to 1.33)
	Colon and rectum	Non-Hispanic White	11.30	10.75	1.05 (1.01 to 1.09)	1.65	1.30	1.27 (1.08 to 1.50)	36.57	35.49	1.03 (0.99 to 1.07)
		Non-Hispanic Black	15.51	9.80	1.58 (1.46 to 1.71)	2.20	1.40	1.57 (1.26 to 1.97)	50.34	31.80	1.58 (1.45 to 1.72)
		Non-Hispanic Asian and Pacific Islander	7.40	7.72	0.96 (0.85 to 1.08)	1.06	1.06	1.00 (0.71 to 1.41)	24.01	25.15	0.96 (0.84 to 1.08)
		Hispanic	9.64	7.63	1.26 (1.19 to 1.34)	1.44	1.11	1.30 (1.11 to 1.51)	31.12	24.71	1.26 (1.19 to 1.34)
	Pancreas	Non-Hispanic White	9.60	9.56	1.00 (0.97 to 1.04)	0.53	0.43	1.24 (0.93 to 1.66)	33.35	33.47	1.00 (0.96 to 1.04)
		Non-Hispanic Black	12.75	8.01	1.59 (1.45 to 1.74)	0.82	0.46	1.78 (1.21 to 2.63)	43.98	27.78	1.58 (1.44 to 1.74)
		Non-Hispanic Asian and Pacific Islander	8.20	6.66	1.23 (1.09 to 1.39)	0.30	0.40	0.75 (0.41 to 1.40)	28.88	23.06	1.25 (1.11 to 1.41)
		Hispanic	7.92	7.84	1.01 (0.95 to 1.07)	0.40	0.35	1.15 (0.86 to 1.53)	27.61	27.44	1.01 (0.95 to 1.07)
	Liver and intrahepatic bile duct	Non-Hispanic White	3.57	3.60	0.99 (0.93 to 1.06)	0.33	0.26	1.27 (0.83 to 1.93)	12.06	12.36	0.98 (0.92 to 1.04)
		Non-Hispanic Black	5.07	4.34	1.17 (1.04 to 1.32)	0.44	0.60	0.74 (0.50 to 1.10)	17.18	14.12	1.22 (1.07 to 1.38)
		Non-Hispanic Asian and Pacific Islander	3.18	5.50	0.58 (0.48 to 0.69)	0.37	0.41	0.91 (0.51 to 1.61)	10.54	18.84	0.56 (0.46 to 0.68)
		Hispanic	5.96	6.05	0.99 (0.92 to 1.06)	0.30	0.32	0.91 (0.67 to 1.24)	20.80	21.05	0.99 (0.92 to 1.06)

^a CI = confidence interval.

All Cancers

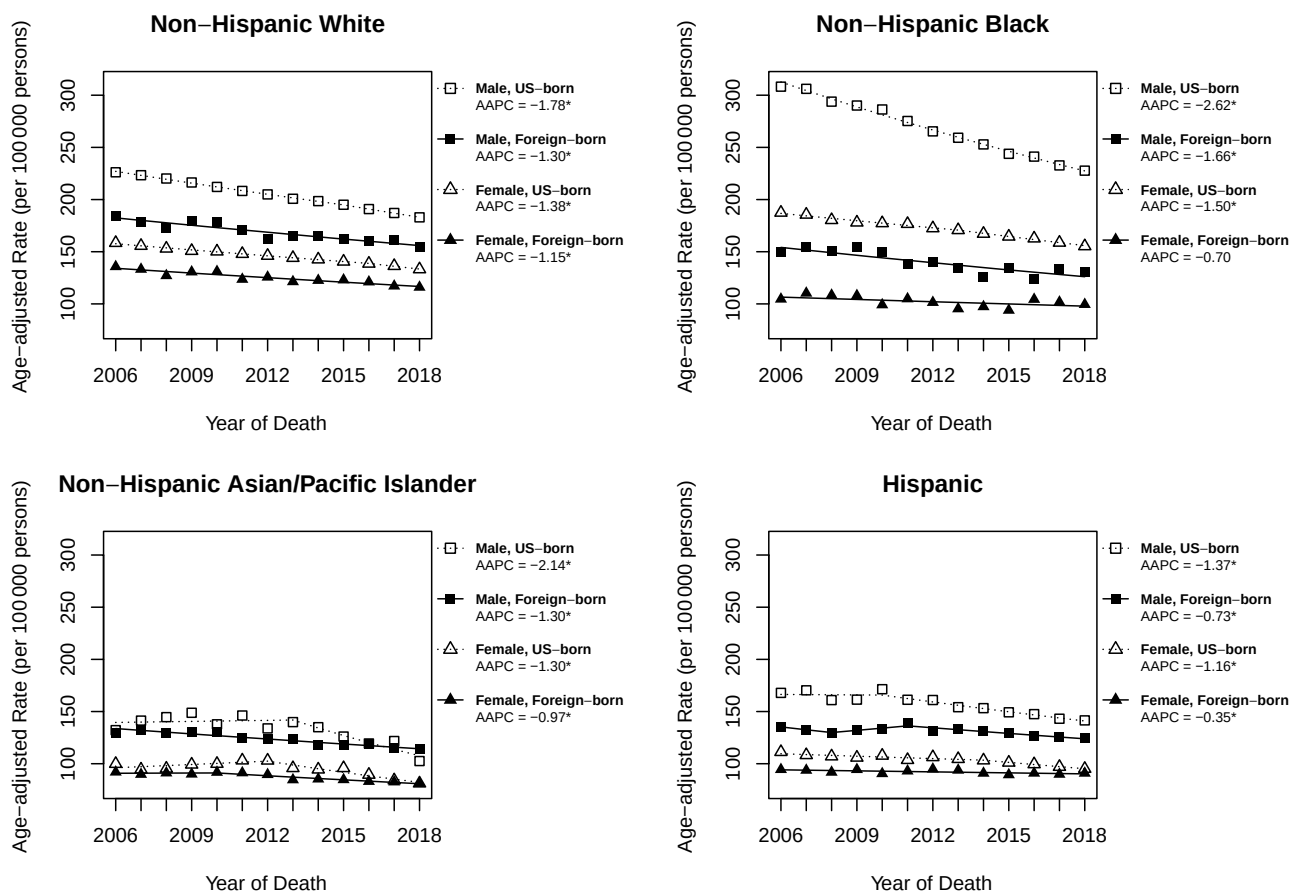


Figure 1. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006-2018 by sex, race and ethnicity, and nativity for all cancers combined. AAPC = average annual percent changes.

nativity, and cancer site. Annual percent change for every segment of a single trendline and average annual percent change across all segments of trendlines (ie, from 2006 to 2018) were computed.

Results

Table 1 shows 3-year (2016-2018) age-adjusted mortality rates for all cancers combined and selected cancers by sex, race and ethnicity, and age group (younger than 50 years and 50 years and older) and respective rate ratios with 95% confidence intervals (CIs). Age-adjusted mortality rates are substantially lower among the foreign-born than their US-born counterparts across almost all cancers and racial and ethnic groups examined, except for the cancer of liver and intrahepatic bile duct for which the reverse is true among Asian and Pacific Islander individuals of both sexes. The largest nativity difference in mortality is observed for lung cancer in Black populations with US-born to foreign-born rate ratios of 3.44 (95% CI=3.18 to 3.72) and 3.67 (95% CI=3.37 to 4.00) for men and women, respectively. US-born Black men also consistently experience statistically significantly higher age-adjusted mortality rates compared with their foreign-born counterparts in other cancers examined with rate ratios ranging from 1.75 (95% CI=1.61 to 1.91) for colorectal cancer to 1.30 (95% CI=1.21 to 1.40) for prostate cancer. The pattern for Black women is similar with US-born to foreign-born rate ratios

ranging from 1.59 (95% CI=1.45 to 1.74) for pancreas to 1.17 (95% CI=1.04 to 1.32) for liver and intrahepatic bile duct cancer. In Hispanic individuals, notably high US-born to foreign-born rate ratios are observed for colorectal cancer in men (rate ratio = 1.38, 95% CI=1.31 to 1.45) and women (rate ratio = 1.26, 95% CI=1.19 to 1.34). US-born Hispanic women also have elevated age-adjusted mortality rates for lung cancer (rate ratio=1.26, 95% CI=1.20 to 1.32). For liver and intrahepatic bile duct cancer, the largest US-born to foreign-born differences are observed in Black men (rate ratio=1.63, 95% CI=1.47 to 1.80) and Hispanic men (rate ratio=1.88, 95% CI=1.79 to 1.99). In Asian and Pacific Islander populations, the largest US-born to foreign-born differences are observed for prostate, female breast, and pancreas. With some exceptions, these patterns hold in the age stratified analysis for the group aged younger than 50 years. The most notable exception is the early onset of colorectal cancer where increases in incidence have been recently recorded in US-born White men (25,26). In our study, an elevated US-born to foreign-born rate ratio of colorectal cancer mortality for White men is observed in the group aged younger than 50 years: 1.46 (95% CI=1.25 to 1.70) compared with rate ratio of 1.01 (95% CI=0.98 to 1.05) in all ages group (Table 1).

Figures 1-7 show the Joinpoint trend analyses of annual age-adjusted mortality rates during 2006-2018 by sex, race and ethnicity, and nativity for all cancers combined and several selected cancer sites. Decreasing trends were observed for US-born and

Female Breast

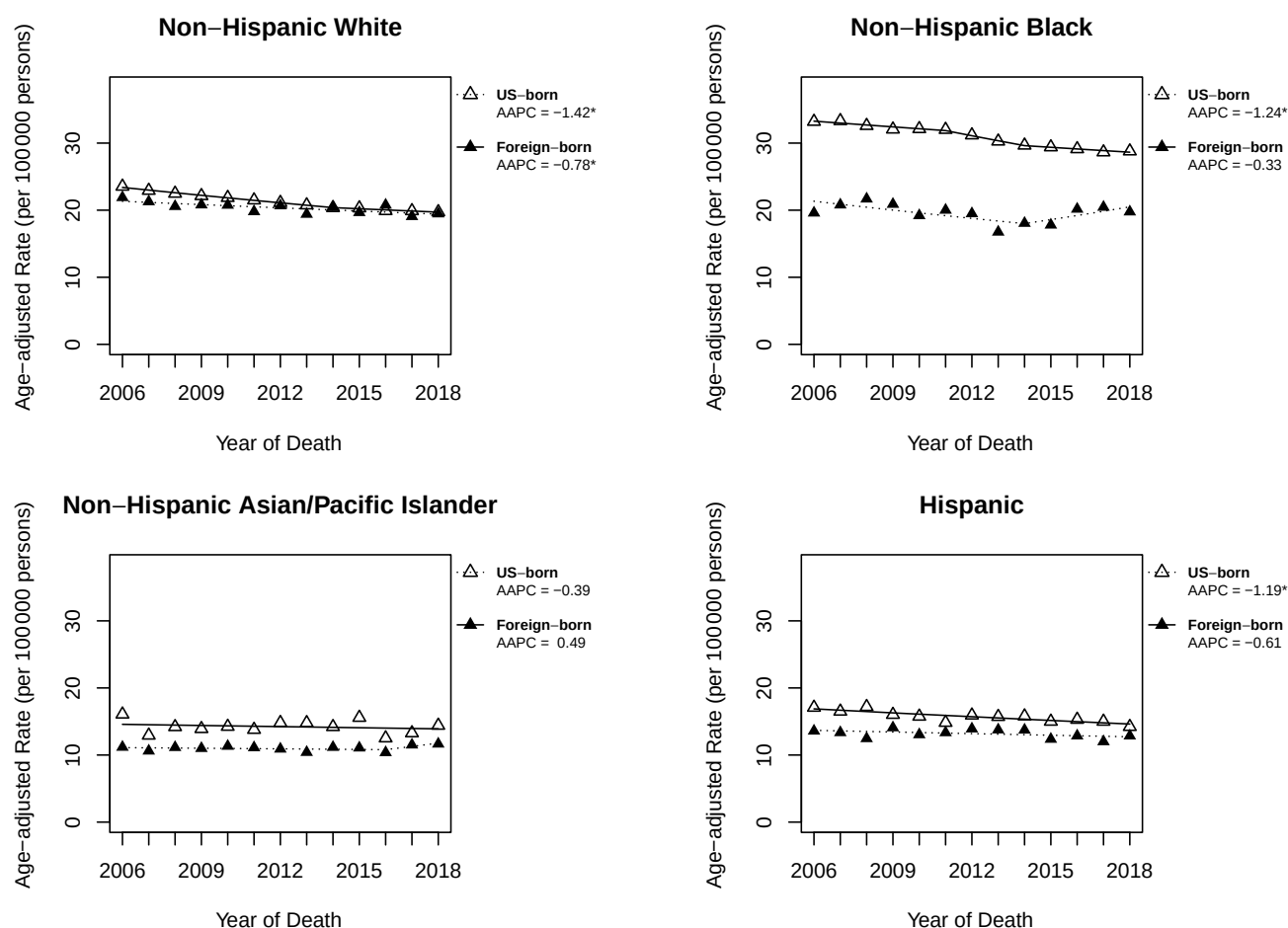


Figure 2. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006–2018 by race and ethnicity, and nativity for female breast cancer. AAPC = average annual percent changes. * indicates that AAPC is statistically different from zero at 0.95 confidence level.

foreign-born groups; however, the decline was more pronounced among US-born groups, as well as White and Black populations of both sexes, and slower among Hispanic and Asian and Pacific Islander populations. Average annual percent changes of all cancers combined for the US-born and foreign-born men were -1.78 (95% CI = 1.82 to -1.73) vs -1.30 (95% CI = -1.61 to -0.97), respectively, in White men, -2.62 (95% CI = -2.78 to -2.45) vs -1.66 (95% CI = -2.45 to -0.70) in Black men, -2.14 (95% CI = -4.11 to -0.08) vs -1.30 (95% CI = -1.50 to -1.07) in Asian and Pacific Islander men, and -1.37 (95% CI = -1.81 to -0.93) vs -0.73 (95% CI = -0.93 to -0.45) in Hispanic men (Supplementary Table 1, available online). Among women, the declining trends were less pronounced than those for men across all nativity and race and ethnicity groups (Supplementary Table 2, available online). Important decreases in mortality were observed mainly for lung, colorectal, and prostate cancers. The steepest decline occurred for lung cancer among US-born Black men (average annual percent change = -4.28 , 95% CI = -4.45 to -4.13) (Supplementary Table 1, available online). However, there were some stable mortality trends and, more importantly, increasing mortality trends for liver and intrahepatic bile duct cancer among US-born White, US-born Black, foreign-born Asian and Pacific Islander, and US-born Asian and Pacific Islander (women only) and Hispanic of both nativity groups. For pancreatic cancer, increasing trends

were observed for women-only US-born White and foreign-born Hispanic (Supplementary Table 2, available online). Although overall declines were more pronounced among groups who had the highest rates in 2006, there was 1 notable exception: breast cancer mortality is decreasing faster in US-born White women than in US-born Black women (Supplementary Table 2, available online).

Discussion

We demonstrated the utility of NCI's SPARC as a publicly available online interactive data mining resource, which is unprecedented, for monitoring population-based cancer mortality by nativity, race and ethnicity, sex, cancer type, and state over time. Our findings have provided in-depth understanding on the impact of nativity on cancer-specific mortality disparities that vary not only by race and ethnicity but also by sex and cancer type and have changed over time corresponding to cancer screening and treatment availabilities.

Using SPARC, the cross-sectional 2016–2018 mortality analysis reveals distinct cancer patterns by nativity in several aggregated racial and ethnic populations in the United States (ie, White, Black, Asian and Pacific Islander, and Hispanic). For all cancer causes combined, US-born populations have higher mortality

Prostate

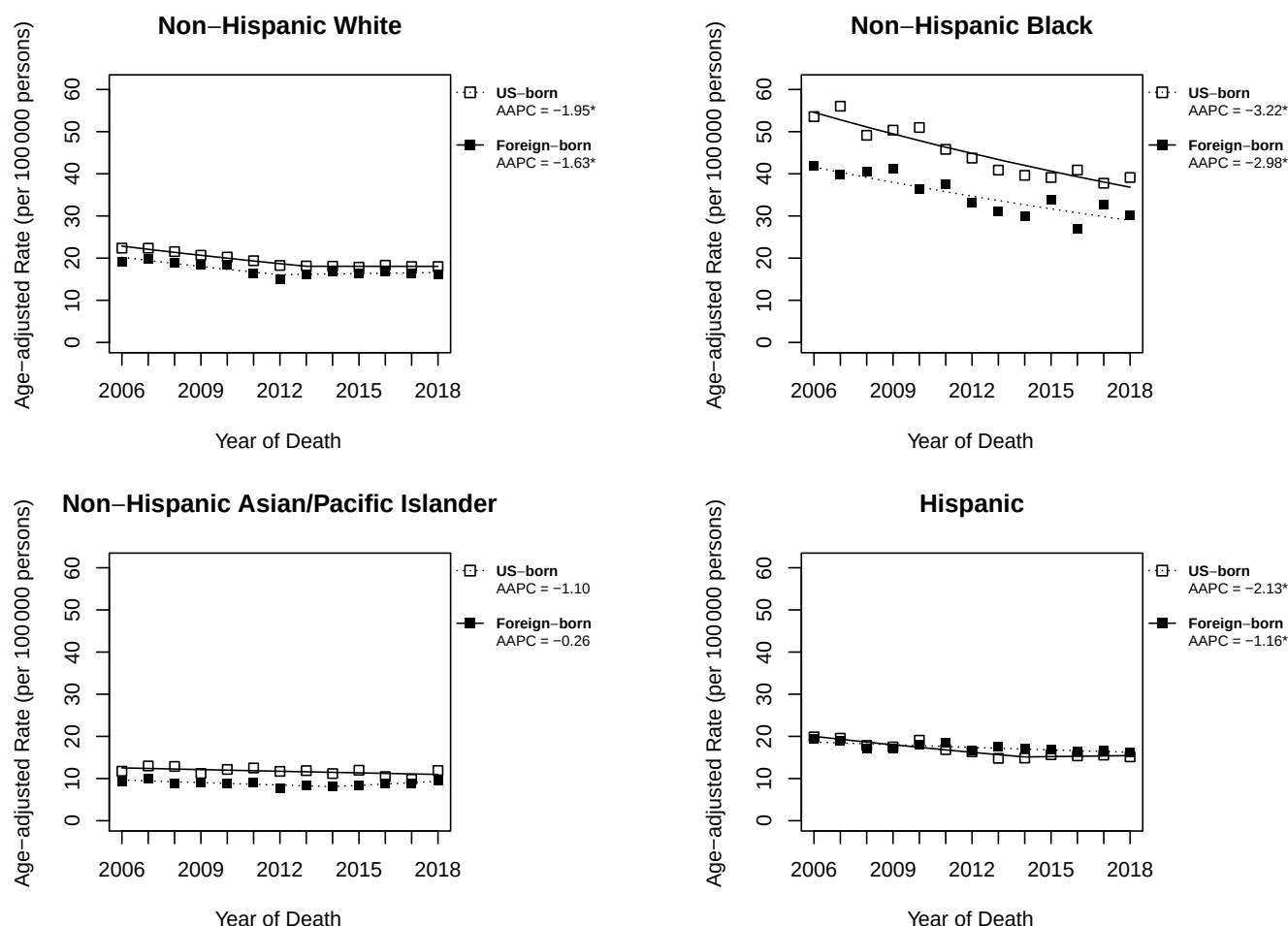


Figure 3. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006–2018 by race and ethnicity, and nativity for prostate cancer. AAPC = average annual percent changes.

than their foreign-born counterparts, and this difference between US-born and foreign-born is particularly pronounced among Black populations, moderate among White and Hispanic populations, and absent among Asian and Pacific Islander populations with variations by cancer site.

The White-Black populations mortality disparities may stem partly from historical discrimination and health-care access barriers (27). However, little attention has been paid to within-race nativity differences. The nearly fourfold higher mortality rates among US-born Black individuals compared with foreign-born Black individuals underscore the impact of tobacco use and obesity among other factors for all cancers. Conversely, low smoking rates in foreign-born Black populations yield the lowest lung cancer mortality across groups (28,29). Despite fewer smoking pack-years, it has been shown that Black individuals exhibit heightened vulnerability to lung cancer death compared with White individuals (30). However, the origin of this vulnerability remains unclear. Foreign-born Black populations may face lesser exposure to unfavorable determinants of health and structural racism especially during formative years than US-born Black populations. However, limited research addresses Black populations like Afro-Caribbean and African immigrants in the United States.

Breast cancer's links to reproductive factors like lower parity and delayed first pregnancy may explain the higher rates among

US-born populations (31,32). However, our study revealed that the key driver of the well-documented White-Black mortality disparity is primarily the excessive rates among US-born Black women in relation to US-born White women, whereas foreign-born Black women's mortality matches that of foreign-born White women. This nativity discrepancy in Black women underscores the need for research into potential genomic and epigenomic differences between US-born and foreign-born Black women and the distribution of aggressive breast cancer types in both populations.

In prostate cancer, higher mortality among Black men is observed in US-born and foreign-born groups, surpassing other groups. Among Black populations, mortality rates are notably higher among US-born compared with foreign-born populations, in line with previous studies conducted in different regions of the United States (5,9). Some research has linked these heightened mortality rates among US-born Black individuals to various adverse factors, including a high-fat diet, obesity, and alcohol consumption, which are more prevalent not only in Western countries but also as an extension among US-born populations (33,34). However, it is advisable to conduct a more comprehensive investigation into the underlying causes of disparities in prostate cancer by nativity group. This examination should encompass incidence rates, which are obviously influenced by

Lung & Bronchus

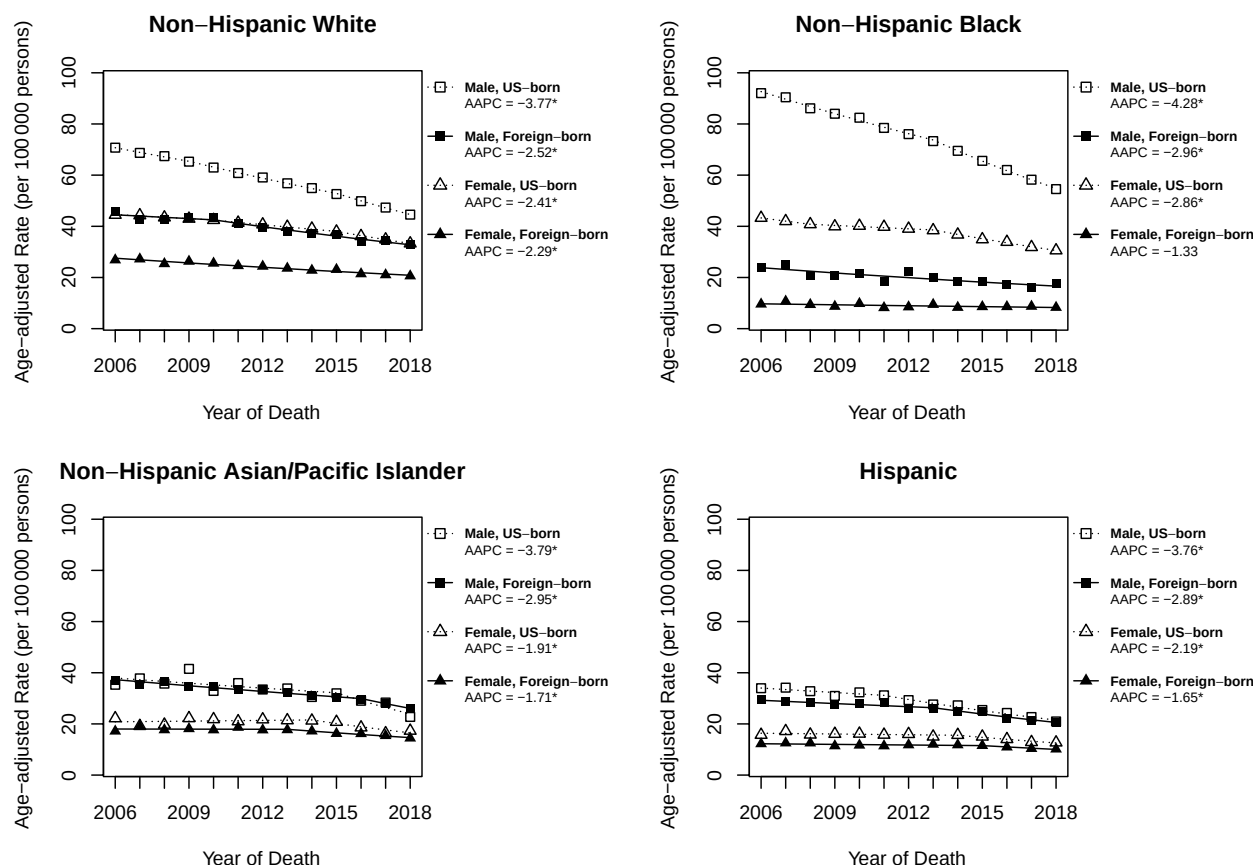


Figure 4. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006–2018 by sex, race and ethnicity, and nativity for lung and bronchus cancer. AAPC = average annual percent changes.

prostate-specific antigen testing, and mortality rates, which are closely associated with access to quality health care and the prevalence of aggressive prostate cancers. Colorectal cancer highlights significant foreign-born advantages, particularly for Black and Hispanic individuals, which likely is linked to obesity, alcohol, and smoking disparities (35,36). For pancreatic cancer, which shared risk factors with colorectal cancer (37), the mortality differences by nativity are mainly observed among Black and Asian and Pacific Islander populations, but the reasons are unclear. Lastly, to understand liver and intrahepatic bile duct cancer patterns, it is crucial to analyze hepatocellular carcinoma by etiology, as hepatocellular carcinoma alone accounts for approximately 80% of all deaths by liver and intrahepatic bile duct cancer. Broadly speaking, hepatitis C virus (HCV)-related hepatocellular carcinoma and alcohol-related hepatocellular carcinoma are more prominent in US-born populations, nonalcoholic fatty liver disease-related hepatocellular carcinoma cases demonstrate a balanced distribution between US-born and foreign-born populations, whereas hepatitis B virus-related hepatocellular carcinoma is more frequent among foreign-born populations, particularly among Black and Asian and Pacific Islander communities (38).

For the first time, we studied mortality trends for the period of 2006–2018 to track cancer progress over time, with a specific focus on race and nativity patterns. This analysis provides essential insights into understanding disparities in the United States and evaluating the progress made in addressing these disparities. Encouragingly, mortality trends show favorable progress with

decreasing rates for all cancer causes combined for every racial group regardless of nativity. The decreasing trend is more pronounced in the US-born compared with the foreign-born populations, which suggests a positive development toward equity, especially considering that these groups had higher age-adjusted mortality rates at the start of the study period. Despite that, circumstantial evidence supporting previously documented phenomena, such as the Health Immigrant Effect (39) and the Hispanic paradox (40–42), which confer a degree of advantage (eg, low mortality rates) among the foreign-born population, persists.

US-born Black individuals recorded the most annual improvement in mortality, including steep decreases for lung cancer followed by prostate and colorectal cancer. Favorable trends for prostate cancer were apparent in all race and ethnicity groups with the strongest gains observed among Black individuals of both nativity groups possibly because of improved treatments. For breast cancer, reductions in mortality were relatively modest and more pronounced in US-born women among all groups except for Asian and Pacific Islander groups. It is unclear if these are because of less favorable incidence trends among foreign-born Asian and Pacific Islander women (32). For lung cancer, gains have been recorded for nearly all groups (except for foreign-born Asian and Pacific Islander), which are possibly linked to decreasing smoking prevalence and increased lung cancer screening with earlier detection (43–45). Colorectal cancer exhibited decreasing trends for all US-born groups (both sexes) and all foreign-born women. Gains were also observed in foreign-born White men and foreign-born Asian and Pacific Islander men

Colon & Rectum – Aged 50 Years or Greater

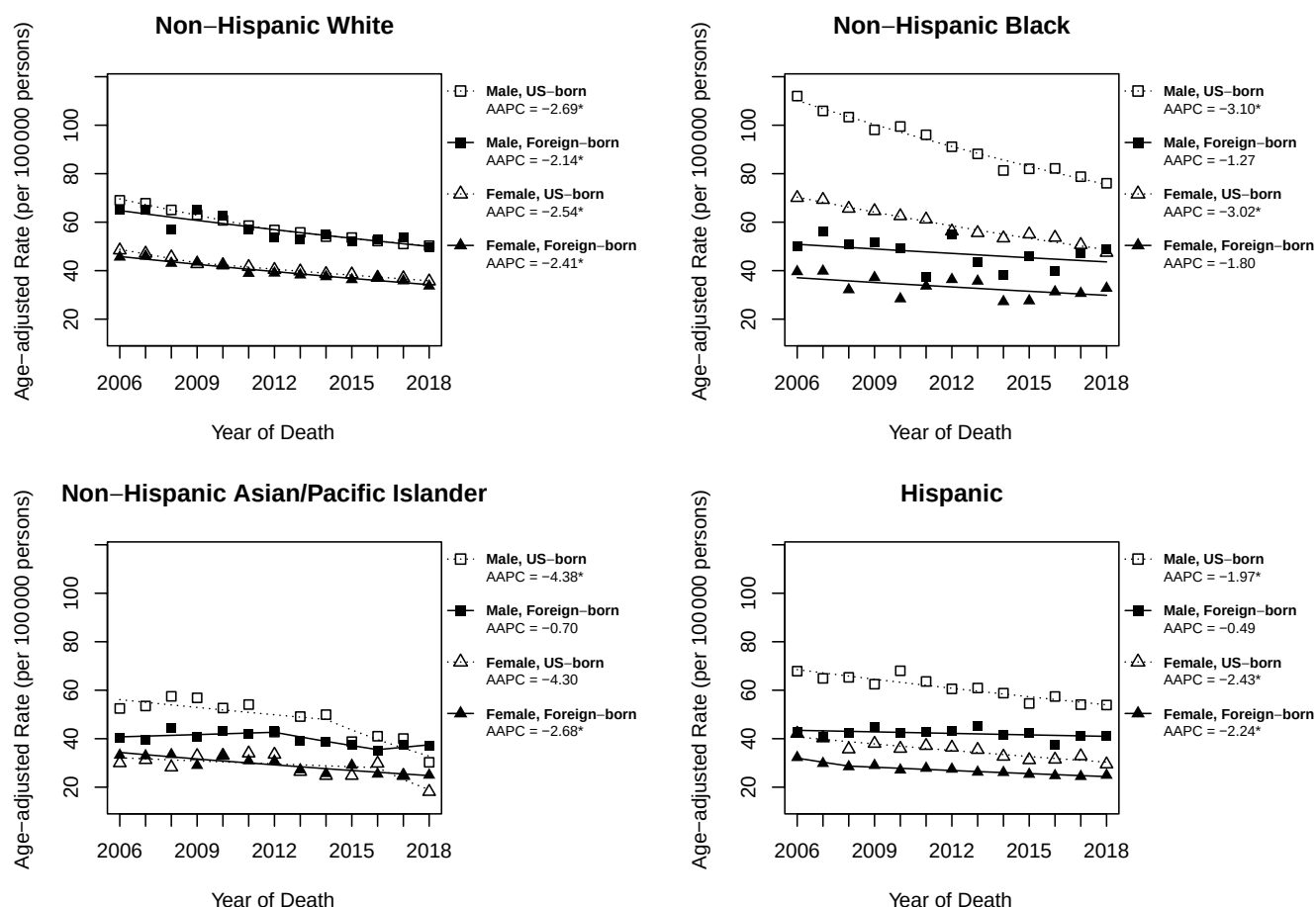


Figure 5. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006–2018 by sex, race and ethnicity, and nativity for colon and rectum cancer. AAPC = average annual percent changes.

but unexpectedly absent in foreign-born Black men and foreign-born Hispanic men. These disparities by sex and race and ethnicity require in-depth evaluation, especially given positive decreases in foreign-born Hispanic women and Black women. Colorectal cancer screening uptake, a key determinant of incidence and mortality, is lower among foreign-born Hispanic and Black men over time, which may stem from “machismo” attitudes and insurance barriers among recently arrived foreign-born populations (less than 15 years) in the United States (46,47). However, insurance would not account for the sex-specific differences. Regrettably, studies on colorectal screening uptake have yet to directly explore sex within analyses of race and ethnicity and nativity.

Not all cancers show decreasing trends. Pancreatic cancer mortality, potentially linked to rising obesity rates (48,49), is increasing slightly among US-born White and foreign-born Hispanic women. Among the various cancers analyzed, liver and intrahepatic bile duct cancer stands out with divergent trends across different demographic groups. Favorable trends observed for foreign-born Asian and Pacific Islander populations are possibly due to hepatitis B virus vaccination efforts in Asia (50). Foreign-born White and foreign-born Black populations show stable trends. However, increasing trends are observed for US-born White, US-born Black, and US-born Hispanic groups. This complex landscape of liver and intrahepatic bile duct trends can

be linked to the different entities that compose liver and intrahepatic bile duct cancer: intrahepatic cholangiocarcinoma, which is on the rise among all racial ethnic groups (51), as well as the cause-specific types of hepatocellular carcinoma, which differ remarkably among each demographic, each cause displaying its own distinct temporal trend (38). In this regard, the recent decrease in HCV-related hepatocellular carcinoma rates, which are higher among US-born populations may not be sufficient to counterbalance the concurrent rise in rates of hepatocellular carcinoma linked to alcohol and nonalcoholic fatty liver disease, resulting in ongoing upward trends observed up to 2018 (38).

This analysis has several limitations. First, the heterogeneity of foreign-born and US-born populations extends beyond nativity, encompassing risk factors and health-care access, which emerges based on country and region of origin, age, education level, socioeconomic status, and life experiences. Japanese individuals account for a substantial proportion of US-born Asian and Pacific Islander individuals, whereas Chinese and Filipino individuals are the top foreign-born Asian and Pacific Islander subgroups (1). Hispanic populations feature Puerto Rican individuals as US-born, whereas Californian foreign-born Hispanic populations are largely Mexican individuals, and Floridian foreign-born Hispanic populations are predominantly Cuban individuals. All these detailed Asian and Pacific Islander and Hispanic subgroups display distinct cancer mortality patterns (52,53). Second,

Pancreas

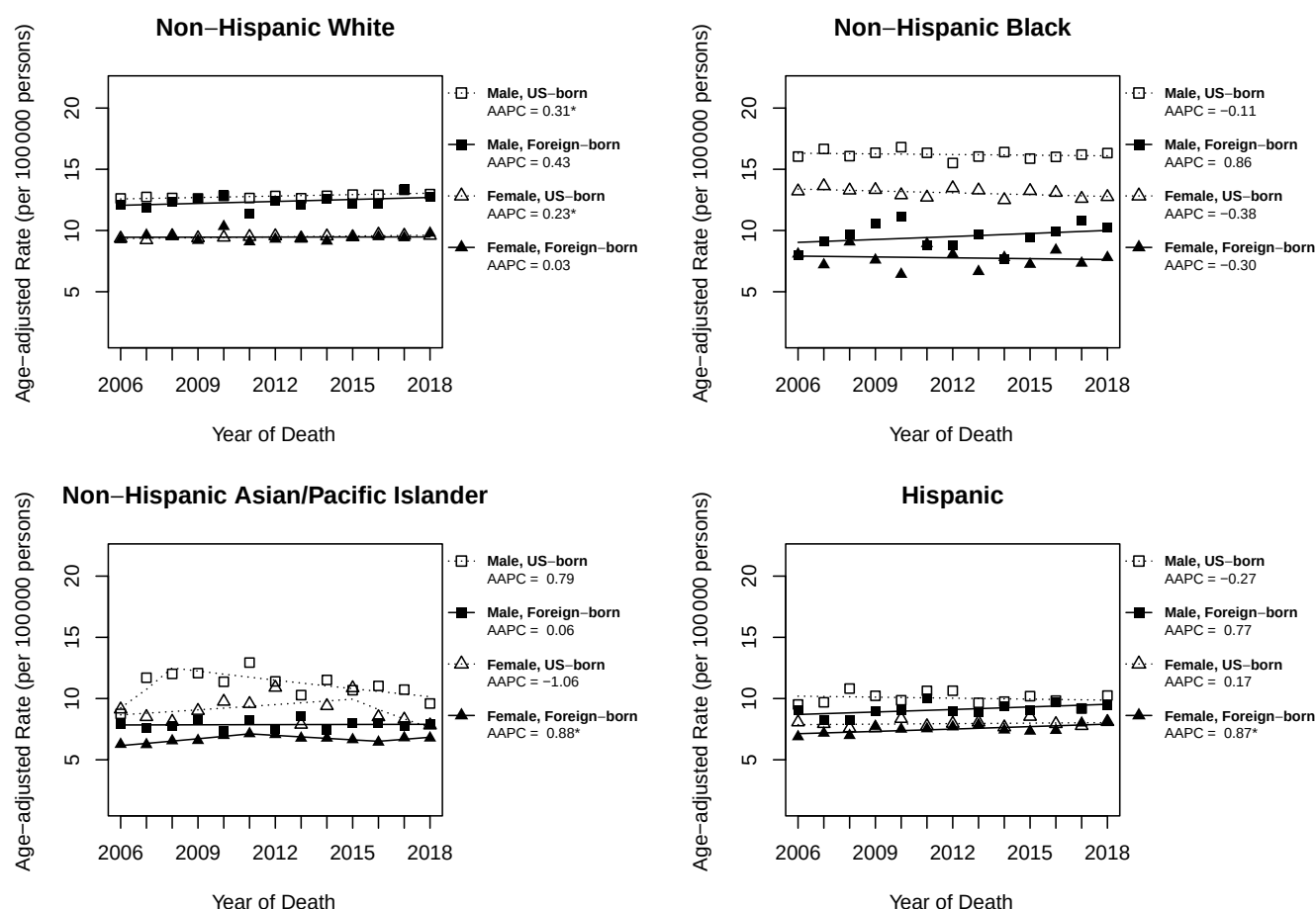


Figure 6. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006-2018 by sex, race and ethnicity, and nativity for pancreatic cancer. AAPC = average annual percent changes.

mortality rates are not always prone to direct interpretation as they encompass not only determinants of cancer incidence, including genetics and cancer risk factors such as smoking and obesity, but also determinants of cancer survival, such as access to and uptake of cancer screenings and cancer treatments. The lack of knowledge regarding these risk factors makes interpretations of cancer mortality results less certain. Third, trends in groups with a large proportion of foreign-born populations, such as Hispanic and Asian and Pacific Islander, will be sensitive to the substantial number of diverse immigrants in recent decades. Fourth, lack of data regarding how long foreign-born individuals have lived in the United States and their ages of immigration prohibits refined analyses to understanding the risk factors that are unique to specific foreign-born subpopulations. Lastly, nativity disparities within the American Indian and Alaska Native populations were not assessed because of the small population sizes in foreign-born American Indian and Alaska Native groups.

Unlike previous reports that used scattered data sources and ad hoc statistical adjustments, this article illustrated SPARC's remarkable contribution to the field of cancer disparity research, particularly by providing a uniform methodology for analyzing comprehensive, nationwide, population-based time series data. This nativity analysis provides unique insights into a crucial indicator of progress against cancer: mortality. Here, we highlight

findings for breast, colorectal, and liver cancer by race and nativity that would have been masked in other analyses that do not account for nativity. Future state-level analyses may shed light on the impacts from state demographic profiles. Cancer mortality varies remarkably between the US-born (higher mortality) and foreign-born (lower mortality) populations that compose current day America. Although the news of descending mortality trends especially among US-born individuals with resultant narrowing disparities are positive, it is important to keep and perhaps intensify the fight against tobacco smoking, obesity, and actionable causes of liver cancer such as chronic infections of HCV and HBV and to increase screening for breast, lung, and colorectal cancer (particularly among foreign-born Hispanic and Black men). Given that most nativity-related disparities occur among Black individuals, conducting studies focusing on US-born Black populations in comparison to foreign-born Black populations is crucial for comprehending the plight of US-born Black individuals regarding cancer mortality.

Data availability

The data for this study are public use data available from the National Cancer Institute's Survey-Based Population-Adjusted Rate Calculator Website (<https://surveillance.cancer.gov/sparc/>).

Liver & Intrahepatic Bile Duct

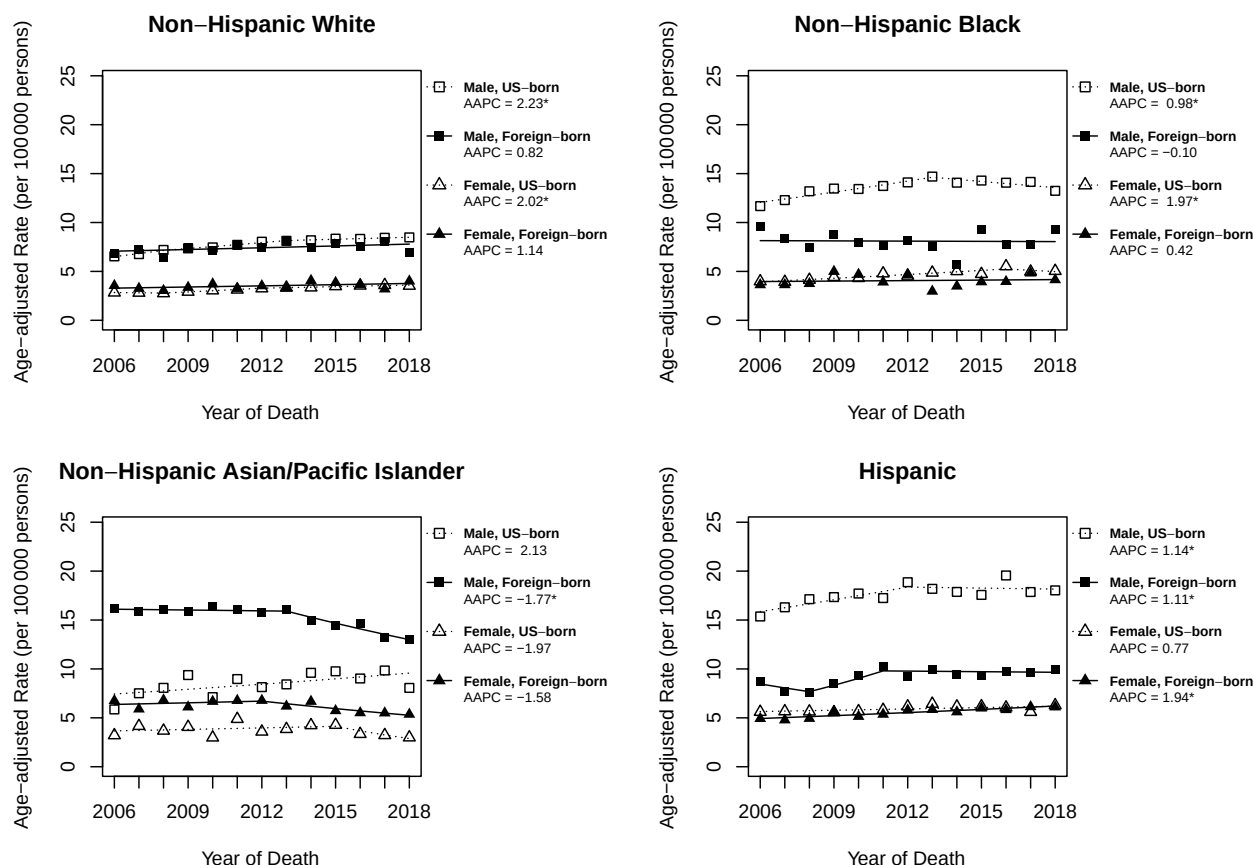


Figure 7. Joinpoint trend analyses of annual age-adjusted mortality rates during 2006–2018 by sex, race and ethnicity, and nativity for liver and intrahepatic bile duct cancer. AAPC = average annual percent changes.

Author contributions

Mandi Yu, PhD (Conceptualization; Formal analysis; Methodology; Software; Supervision; Visualization; Writing—original draft; Writing—review & editing), Lihua Liu, PhD (Conceptualization; Methodology; Writing—original draft; Writing—review & editing), James Todd Gibson, BA (Data curation; Formal analysis; Methodology; Writing—review & editing), David Campbell, BA (Formal analysis; Software; Writing—review & editing), Qinran Liu, MA (Validation; Writing—original draft; Writing—review & editing), Steve Scoppa, BA (Conceptualization; Data curation; Methodology; Software), Eric J Feuer, PhD (Conceptualization; Methodology; Software; Supervision; Writing—original draft; Writing—review & editing), and Paulo S. Pinheiro, PhD (Conceptualization; Formal analysis; Investigation; Methodology; Supervision; Writing—original draft; Writing—review & editing)

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Conflicts of interest

No conflict of interest was reported by any of the authors.

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